# Supplementary Material: ODD Description of the Agent-Based Artificial Economy

for "Mechanism-Oriented Agent-Based Modeling (MO-ABM): A Methodology for Identifying the Indispensable Model Structures and Causal Mechanisms of Socio-Economic Phenomena"

Shigeaki Ogibayashi

Professor Emeritus, Chiba Institute of Technology, Japan · ogibayashi@ogi-lab.net

This document describes the artificial economic model used in Sect. 4 of the paper (C–S, C–R–W, C–R–W–S, and C–S–G systems) following the ODD protocol (Grimm et al., 2020). Equation and table numbers refer to the paper. The C++ source code is available at https://github.com/ogi-lab/MO-ABM-CSS2026; release v1.1-CSS2026 was used for Fig. 3 and release v1.0-CSS2026 for Figs. 4–6 and Table 2 (see Sect. 8 below).

**Table 1** ODD description of the economic model

| ODD element | Description |
|---|---|
| 1. Purpose and patterns | Purpose: to identify the model structures indispensable for reproducing (a) price equilibrium, (b) positive profits of all producer categories and stable GDP without external funds (an autonomous economy), and (c) a GDP increase caused by an income tax cut, in an economy with autonomous money circulation, and to show that parameters affect these qualitative characteristics only as thresholds. Patterns (qualitative characteristics) used for evaluation: (a) production, purchases, average market price, and GDP converge to values independent of the initial price; (b) the total profit of R and W remains positive without external funds; (c) the income tax rate and GDP are negatively correlated. |
| 2. Entities, state variables, and scales | Entities: consumers (C), retailers (R), raw-material makers (W; "Wholesaler" in the code), sole proprietors (S), and the government (G). State variables of producers (R, W, S): price p, production quantity q, stock s, maximum production $q_{max}$, target stock $q_s$, cash and deposits, and the accounts of the balance sheet and income statement. State variables of consumers: income I, budget $E_b$, utility weight $w_k$ for each good, and deposits. Government: income tax rate, market purchase ratio mpr, and tax revenue. The model has no spatial structure. One step represents one accounting period; it corresponds to one month in models that include corporate investment, but in the present model, which has no investment, it is not mapped onto a specific calendar period. Simulations run for 150–800 steps. |
| 3. Process overview and scheduling | Each step is processed in the following order: (1) producers determine production quantity by Eq. (2) and shipping price by Eq. (3) and produce; the market (market class) compiles the list of goods in stock, which all producers and consumers can refer to; (2) firms withhold income tax and pay it to the government, and pay after-tax wages and executive compensation (wages depend on the previous period's profit and equal the base wage in the initial period; the profit of sole proprietors becomes their income); (3) consumers purchase goods according to the rule of Eq. (1); (4) the government purchases goods with mpr × total tax revenue; (5) the market compiles the end-of-period stock list, each agent closes its accounts by double-entry bookkeeping, and the input–output table and GDP of the whole system are computed. The order of processing between agent types is fixed; within each type, agents are processed in a random order that is reshuffled every step. |

| ODD element | Description |
|---|---|
| 4. Design concepts | Basic principles: macro phenomena emerge bottom-up from micro decision rules. The content and order of processing are made as close as possible to those of real economic systems. GDP and the input–output table are computed endogenously from the double-entry bookkeeping of each agent by essentially the same method as in national accounts and input–output statistics (real statistics require estimation of missing values, whereas in the model all transactions are recorded). Emergence: price equilibrium, positive profits, the level and time course of GDP, and the relation between tax rate and GDP. Adaptation: producers adjust production and price according to stock and sales (Eqs. (2) and (3)). Objectives: consumers buy, one unit at a time, the item with the largest utility of one more unit under the budget constraint (greedy maximization of a utility with diminishing marginal utility). Learning and prediction: none (myopic adjustment based on information from previous periods). Sensing: producers know their own stock on the market, their sales over the last n periods, and the average market price (or the previous period's average transaction price); consumers know the goods on the market and their prices. Interaction: indirect, through trade in the goods market and through employment and wages. Stochasticity: utility weights $w_k$ (uniform in [0, 1]) given to each agent; initial price, minimum price $p_{min}$, base wage, and initial cash and deposits (each uniform within a specified range) given to each agent; dispersion of sole proprietors' shipping prices (±6%); bonus ratio of employees (uniform in [0.05, 0.07] each period); random choice among goods of equal price; and the processing order of agents within each type. Shipping prices are set by each producer before shipment, whereas the market price is a system variable determined through interaction with consumers. Collectives: none. Observation: GDP, total profit of R and W, average market price, and total production; in the income tax experiments, the mean GDP over the last 100 steps. |
| 5. Initialization | Numbers of agents are given in Table 3 of the paper. Initial prices: the four ranges of Table 3 (uniform random within each range, given to each agent). Initial stock: 0. Initial cash and deposits: uniform random within a specified range for each agent, set sufficiently large compared with the wage level (changes during the simulation are small compared with the initial values, so their values hardly affect the results). Base wages: uniform random in 2,500–4,000 for each agent. Baseline random seed: 500. |
| 6. Input data | None (the model is fully endogenous). Parameters are read from the files Set_parameter_1.csv to Set_parameter_9.csv; their contents are described in the README of the code repository. |
| 7. Submodels | Consumer's purchase: the procedure of Eq. (1) with $\rho = -2$; an item that is out of stock or unaffordable is excluded from the candidates for the period (release v1.1). Production: Eq. (2) with stock ratio r, adjustment rate $\beta$, target stock $q_s = \mu + \delta\sigma$ computed from sales over the last n = 10 periods ($\delta = 1.65$), carry-over of rounding remainders (release v1.1), and bounds $q_{min}$ and $q_{max}$. Pricing: Eq. (3) with adjustment rate $\alpha$ and lower bound $p_{min}$; shipping prices of sole proprietors vary uniformly by ±6%, those of R and W do not vary. Choice of goods: the cheapest unit of the chosen item, at random among equal prices. Baseline: $r = 0$, $\beta = \alpha = 0.05$. Wages: the base wage of each employee and executive is uniform random between the lower and upper bounds (2,500–4,000); each period the wage paid is the base wage plus rate × (profit − base wage). Because this addition applies regardless of the sign of profit, the wage paid is below the base wage in periods in which profit is below the base wage. The rate is a bonus ratio, drawn each period from [0.05, 0.07] for each employee and fixed at 0.2 for all executives; wage_ratio multiplies the base-wage range (e.g., 3,750–6,000 for wage_ratio = 1.5). Income of sole proprietors: their profit becomes household income directly. Consumption function: budget $E_b = c_{min} + rpc \cdot I$, where I is after-tax income, $c_{min} = 100$, and rpc is the marginal propensity to consume (0.7 or 0.8 in the income tax experiments, 1 otherwise). Government: income tax is withheld before wages and executive compensation are paid and remitted by firms (sole proprietors likewise withhold and remit tax from their income); the government purchases goods on the market with mpr × total tax revenue using the consumers' purchase rule, and the remaining revenue is not spent. |

| ODD element | Description |
|---|---|
| 8. Code versions | Release v1.1-CSS2026 (used for Fig. 3) implements the submodels as described above, except for two minor points that also apply to v1.0: in the pricing rule, the condition on production is evaluated with the production quantity already updated for the next period ($q_{i+1}$) instead of $q_i$ in Eq. (3); and for $r > 0$ the second condition of Eq. (2) is implemented as $s_i \geq \bar{s}_i$. These points are not expected to affect the qualitative results and will be corrected in a later release. Release v1.0-CSS2026 (used for Figs. 4–6 and Table 2) differs in the following respects: when a consumer encounters an unaffordable item, purchasing for that period ends instead of continuing with other items; and production quantities are rounded without carrying the remainder over to the next period; in addition, the end-of-period stock records of sole proprietors are registered in the data analyzer of the raw-material market instead of that of the retail market. |

## Reference

Grimm, V., Railsback, S. F., Vincenot, C. E., et al. (2020). The ODD protocol for describing agent-based and other simulation models: A second update to improve clarity, replication, and structural realism. *Journal of Artificial Societies and Social Simulation*, 23(2), 7.